

Camara - 12P

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
II SEMESTER 2018-2019
EEE/CS/INSTR F241 MICROPROCESSOR PROGRAMMING AND INTERFACING
Comprehensive Exam (OPEN BOOK)

M.M: 80

03-05-2018

DURATION: 2hrs

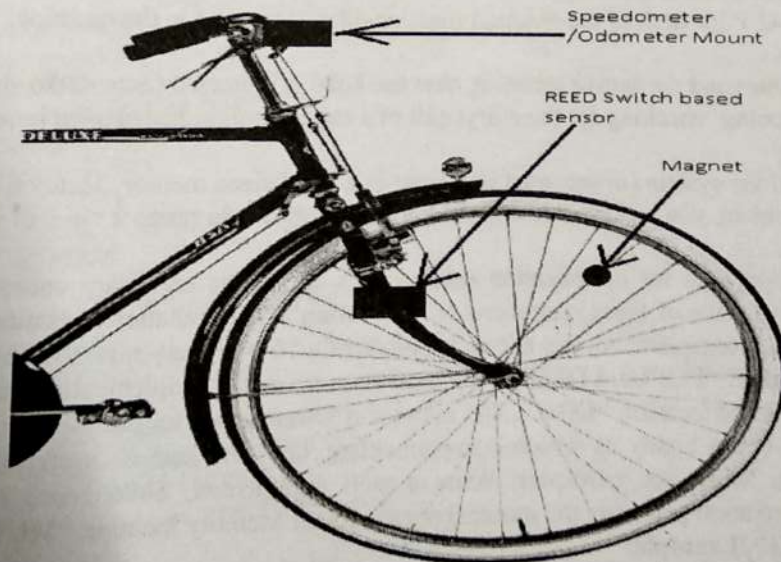
Note: Attempt all parts of the questions at one place only

Q1. Design a bicycle odometer using 8086 microprocessor. Odometer keeps a record of total distance travelled by the bicycle (in Km). There is a magnet and magnetic reed switch is mounted on the wheel spokes and wheel fork respectively as shown in the illustration. Each time magnet passes through the wheel fork, magnetic reed switch closes, which generates a low to high pulse which is given to IR1 input of 8259A programmable interrupt controller (PIC). A distance of 100 meters is covered on 50 rotations of the bicycle wheel. Mask all interrupts other than IR1. Assume IR0 interrupt type is mapped to 90H.

- 91H
- Write ICW register values required in this interrupt based application.
 - Write OCW values required in this application.
 - Write ALP to initialize the interrupt controller using ICW and OCW.
 - Write ISR for the given interrupt type to count number of pulses and increment the distance Value by 100 meters. The distance in Meter is stored in word size memory location named MTR and distance in kilometer is stored in word sized memory location Named KM. (ISR must exit with non specific end of interrupt (EOI) command).
 - Draw the schematic diagram of 8259A interfacing with 8086. It should clearly show address decoding and other required connections between processor and PIC. Assume 8259A base address as 740H.

Note: No need to interface Display device and its code snippet.

[20M]



Q2. An 8086 microprocessor based system requires 32KB of ROM and 64 KB of RAM memory. The Size of Available ROM and RAM chips are 16 KB and 32KB respectively. Assume both types of memories are byte addressable. You are supposed to select the starting address of ROM suitably and the RAM address must start at 00000H. Assume contiguous memory locations for both ROM/RAM.

a). Write the entire range (memory map) of RAM/ROM addresses using the format given in the table.

b). Draw the complete labelled memory interfacing diagram using absolute addressing. use only NAND gates if required (you can choose variable multiple input NAND gates). Show all control signals. [20M]

Memory Chip	(Start Address in Hexadecimal)	(End Address in Hexadecimal)

Q3. You are supposed to design a 8086 based Smart Car parking lot system. The working of the system is as follows. There is a display board outside the parking lot which displays the number of vacant slots in the car park. At the entry/exit of the car park there are individual sensors which detects the entry/exit of a car. Also there are individual UID scanners at the entry and exit. As soon as a car enters, the UID (unique ID) of the car is scanned by a device and its entry time (T_{IN}) is noted. Once the car enters the parking, the available slot count on the display board is reduced by 1. Similarly, when a car exits, the UID and exit time (T_{OUT}) is noted and the vacant slot count is incremented by one. The system maintains a real time clock in seconds which is used for recording the entry/ exit time. Assume that the 8-bit UID value of an incoming car is given by a parallel input device. Based on the duration of the stay, the money to be charged is evaluated as $C_{sec} * (T_{OUT} - T_{IN})$. The C_{sec} is the cost of stay per second. The money to be charged from individual users are stored in the system. No need to display the charge.

Following components are given:

8086 - 1 No., 8255- 1 No., seven segment displays - 2 Nos., 7447- 1 No., logic gates (as per need).

a). Design the smart parking lot system using the given components. Show the interfacing diagram with 8086. For each of the chips (e.g. 8255) clearly highlight the mode, control words, etc. The base address of 8255 is 87H. If you require to use INTR interrupt use interrupt type 64H. However, there is no need to implement the circuitry to send this interrupt type to 8086.

b). Write the ALP for each of the required functionalities described in the question.

Constraints:

When a car comes and the sensor senses it, then the 8086 is intimated (note: 8086 should not be constantly scanning/ checking for the entry/ exit of a car as it might be engaged in some other tasks).

Assumptions:

Current time of the system (in seconds) is present in a word-sized memory location called TIME. For solving the problem, you can assume that the number of slots is in range $0 < \text{no. of slots} < 99$. [30M]

Q4. Genetic algorithm for optimization uses various operations on binary coded values to optimize certain functions. One of these operations is "Mutation". The mutation operation is implemented in such a way that it complements bits at two locations of a 16-bit binary number. The 16 bit numbers are stored at a location "POPULATION". The locations, at which complementing has to be performed is stored at bytes sized location "LOC". The upper and lower nibble for each data at LOC represents the two locations (from right) at which complementing has to occur (0 corresponds to LSB and F corresponds to MSB, see example). Write a well commented, 8086 based ALP for performing MUTATION operation and store the mutated population at Memory location "MUTATED". (The size of populations is 30) Example: [10M]

POPULATION	LOC	MUTATED
2156H (0010 0001 0101 0110)	19H	2354H (0010 0011 0101 0100)

Voltage Amplifier
Current Amplifier
Transconductance Amplifier
Resistance Amplifier